WORKSHEET 3.6

Second and Higher Derivatives

Unit 3: Composite, Implicit and Inverse Functions

Key ideas

- The second derivative is the derivative of the first: $f''(x)$, or $\dfrac{d^2y}{dx^2}$.
- For motion: position s, velocity s', acceleration s''.
- Implicitly: differentiate $\dfrac{dy}{dx}$ with respect to x (quotient rule), then **substitute** for $\dfrac{dy}{dx}$.
- Repeated derivatives of sin, cos and e^{kx} follow patterns: use the pattern, do not differentiate fifty times.

Worked examples

Example 1 · A polynomial

Find $f''(x)$ for $f(x) = x^4 - 2x^3$.

Solution. $f'(x) = 4x^3 - 6x^2$.

$f''(x) = 12x^2 - 12x$**.**

Example 2 · A pattern

Find the third derivative of $y = e^{3x}$, and a formula for the nth derivative.

Solution. Each derivative multiplies by 3.

$27e^{3x}$**; in general** $3^n e^{3x}$**.**

Example 3 · Implicitly

Find $\dfrac{d^2y}{dx^2}$ for $x^2 + y^2 = 9$.

Solution. $\dfrac{dy}{dx} = -\dfrac{x}{y}$. Then $\dfrac{d^2y}{dx^2} = -\dfrac{y - x\frac{dy}{dx}}{y^2} = -\dfrac{y + \frac{x^2}{y}}{y^2} = -\dfrac{x^2 + y^2}{y^3}$.

$-\dfrac{9}{y^3}$**.**

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Practice

CORE

1. Find $f''(x)$ for $f(x) = 5x^3 - x$.

2. Find $\dfrac{d^2y}{dx^2}$ for $y = \cos(2x)$.

3. Find $\dfrac{d^2y}{dx^2}$ for $y = \ln x$.

4. A particle has position $s(t) = t^3 - 3t^2$. Find its acceleration.

EXAM LEVEL

5. Find $f''(x)$ for $f(x) = x^2e^x$.

6. Find $\dfrac{d^2y}{dx^2}$ for $y = \sin^2 x$, in simplified form.

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7. For the curve $y^2 = 4x$, find $\dfrac{d^2y}{dx^2}$ at the point $(1, 2)$.

8. Find the fourth derivative of $y = \sin(2x)$.

Question 9 · Free-response practice

Consider the curve $x^2 + 2y^2 = 6$.

(a) Find $\dfrac{dy}{dx}$ in terms of x and y.

(b) Show that $\dfrac{d^2y}{dx^2} = -\dfrac{3}{2y^3}$.

(c) Find the values of $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ at the point $(2, 1)$. Is the curve concave up or concave down there?

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Worked solutions

1. $f'(x) = 15x^2 - 1$, so $f''(x) = 30x$.

2. $y' = -2\sin(2x)$, so $y'' = -4\cos(2x)$.

3. $y' = \dfrac{1}{x}$, so $y'' = -\dfrac{1}{x^2}$.

4. $s'(t) = 3t^2 - 6t$, so $a(t) = 6t - 6$.

5. $f'(x) = (x^2 + 2x)\,e^x$, so $f''(x) = (x^2 + 4x + 2)\,e^x$.

6. $y' = 2\sin x \cos x = \sin(2x)$, so $y'' = 2\cos(2x)$.

7. $2y\dfrac{dy}{dx} = 4$, so $\dfrac{dy}{dx} = \dfrac{2}{y}$ and $\dfrac{d^2y}{dx^2} = -\dfrac{2}{y^2}\dfrac{dy}{dx} = -\dfrac{4}{y^3} = -\dfrac{1}{2}$.

8. Each pair of derivatives multiplies by -4: $16\sin(2x)$.

9. (a) $2x + 4y\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{x}{2y}$.

   (b) $\dfrac{d^2y}{dx^2} = -\dfrac{2y - 2x\frac{dy}{dx}}{4y^2} = -\dfrac{2y + \frac{x^2}{y}}{4y^2} = -\dfrac{2y^2 + x^2}{4y^3} = -\dfrac{6}{4y^3}$.

   (c) $\dfrac{dy}{dx} = -1$ and $\dfrac{d^2y}{dx^2} = -\dfrac{3}{2}$. The second derivative is negative, so the curve is concave down.

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